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Code No: CS350 USN:

JSS Mahavidyapeetha
Sri Jayachamarajendra College of Engineering (Autonomous)
Mysuru – 570 006
Under Visvesvaraya Technological University, Belagavi

III Semester B.E. Degree
Semester End Examination, December 2016

Branch : CS & E Section: A & B

Name of the Paper Setter: Sheela. N and Manimala. S

DATA STRUCTURES

Duration: 3 Hours Time: 9.30 to 12.30 pm Max. Marks: 100

Date: 19.12.2016 Day: Monday

1. Part – A is compulsory
2. Part – B has internal choice.

Course Outcome:	
CO–1	Understand abstract data structures and perform operations on stack.
CO–2	Develop applications using queue data structures.
CO–3	Implement various operations on linear list.
CO–4	Comprehend the concept of non-linear data structure.
CO–5	Create an application using BST and understand basic operations on graph.

PART – A

Q. Nc	CO	Cognitive Domain	Question	Marks
1(a)	CO1	Knowledge	Define data structure. What are its goals? Give any four examples.	06
(b)	CO1	Knowledge	Explain abstract data type with examples.	04
2	CO2	Synthesis	With complete implementation details propose an efficient alternate method of implementing ordinary queue to overcome its draw backs.	10

3(a)	CO3	Analysis	Appraise the advantages of linked representation over array representation of list of data items.	05
3(b)	CO3	Evaluation	Design an algorithm that traverses a circular single linked list of integers and deletes all the nodes following a node with negative value if exists.	05
4(a)	CO4	Comprehension	Illustrate with an example the different storage representation of a binary tree.	05
(b)	CO4	Comprehension	What are the total number of nodes and total number of leaf nodes in a complete binary tree of depth'd' prove tne same.	05
5(a)	CO5	Analysis	With examples discuss how threaded binary trees are useful in resolving the drawbacks of binary tree.	06
(b)	CO5	Synthesis	Design an algorithm that finds the sum of degrees of a node in an undirected graph using the adjacency list representation.	04

PART – B

6(a)	CO1	Application	Write an algorithm to convert infix expression to postfix expression	05
(b)	CO1	Synthesis	Applying the required algorithm determine the value of the following postfix expressions for A = 2, B = 3, C = 4 and D = 5 by tracing the stack contents. i) A B C D * - + ii) D C * B A + -	05
			OR	
7(a)	CO1	Evaluation	Using only stack ADT, design a routine to pop out a largest element in the stack of integers retaining the order of the remaining elements	05
(b)	CO1	Synthesis	Change the following infix expression to postfix form applying the algorithm and tracing the stack contents i) (A+B) * C - D * F + C ii) (A – 2 * (B + C) – D * E) * F	05

8(a)	CO2	Analysis	A circular queue of size 5 has three elements 15, 25, 35 which are inserted in the same sequence. After insection Rear=4. Show the contents of the queue along with front and rear index values with circular representation for the	05

			following sequence of operations i) Insert 50 and 60 ii) Insert 70 iii) Delete two elements iv) Insert 100	
(b)	CO2	Application	Discuss with an example, application of queue in categorizing items i.e to group the numbers maintaining their original order.	05
			OR	
9(a)	CO2	Evaluation	Using only queue ADT, and array representation write an algorithm called Cat - Queue that concatenates two queues together. The second queue should be put at the end of the first one.	05
(b)	CO2	Analysis	List and write the routines to implement input restricted double ended Queue by defing queue structure.	05
10	CO3	Application	Discuss with an example linked list representation of polynomials write a routine called ADD (P1, P2, P3) which adds two polynomials (2 variables) pointed by P1 and P2 returns the resulted polynomial in P3. Mention the type of list used for representation define the structure for nodes and write initialization statements to initialize the list.	10
			OR	
11	CO3	Application	Write routines to perform the following operations using circular double linked list with header node to handle list of integers. Define node structure and initialize empty list. i) To insert a node by position if position exists ii) To search for a node with the information X and updates it by Y provided X exists in the list iii) To delete the front most node and insert it to the end of the list.	10
12	CO4	Comprehension	Construct the expression trees for the following expression and write the other two forms of the expression by traversing in the required order. i) $ - * + A B C – D E + F G ii) A B + C * E $ F G * /	10

			OR	
13(a)	CO4	Comprehension	Explain the following with examples i) Depth and height of a node in a tree ii) Complete binary tree	04
(b)	CO4	Analysis	Design the routine for iterative preorder traversal and trace the same with a complete binary tree having six nodes.	06

14	CO5	Analysis & application	Construct the binary search tree for the sequence of inputs 80, 50, 30, 90, 95, 65, 40, 20, 75, 62, 85, 88, 94, 63, 59, 60. Discuss how deletion operation is preformed while deleting 50, 95, 65, 80 in sequence. Write the tree after each deletion.	10
			OR	
15	CO5	Evaluation	For the graph shown in fig.1 give DFS and BFS traversals starting from vertex A by tracing the contents of the required data structures. Use adjacency matrix representation.	10

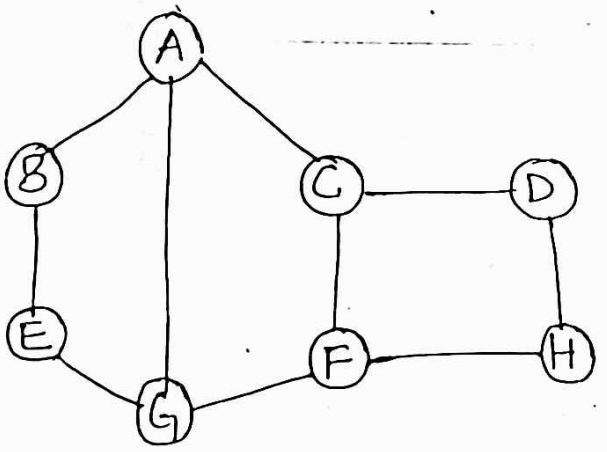


Fig. 1.

Question paper
JSS SCIENCE AND TECHNOLOGY UNIVERSITY

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Course: CS340-A - DATA STRUCTURES Event: 18 UG ODD

Code No: CS340

JSS MAHAVIDYAPEETHA
JSS Science and Technology University
Mysuru - 570 006

III Semester BE Degree
Semester End Examination - December 2017

Branch : Computer Science & Engineering **Section: 'A' & 'B'**
Name of the Paper Setter: Manimala S.

DATA STRUCTURES

Duration: 3 Hours **Time: 9.30 am to 12.30 pm** **Max. Marks: 100**
Date: 14.12.2017 **Day: Thursday**

Instructions to the Candidates:

Answer all questions.

Course Outcomes	
CO–1	Understand and analyze the fundamentals of data structure and their application essential for problem solving.
CO–2	Analyze Linear Data Structures State, Queue and develop applications using the same.
CO–3	Implement various operations on Linear list.
CO–4	Comprehend the concept of non-linear data structure Tree and develop applications.
CO–5	Implement priority queues for various applications and use nonlinear data structure graph for different applications.

PART – A

Q. No	CO	Cognitive Domain	Question	Marks
1.a)	CO1	Application	Write a function in C to find the frequency of the character using the prototype below int Frequency char (char & str, char ch); where str is the input string and ch is the character whose frequency is to be determined.	05
b)	CO1	Knowledge	Discuss the concept of ADT.	05
2.a)	CO2	Application	Write an algorithm to convert the given infix expression to postfix expression. Convert the expression $((A+B-C)*D)\$(E-F/G)$ into its postfix equivalent using stack.	08
b)	CO2	Knowledge	Enumerate linear data structures.	02

3			Implement the Queue ADT using circular singly linked list. Write the main function to demonstrate the queue operations. Show the status of queue. insert (10), insert (20), delete ( ), delete ( ), insert (30), insert (40), insert (10).	10
4	CO4	Application	Construct the BST for the data 14, 10, 17, 12, 11, 20, 18, 25, 8, 22, 23. Show the BST after deleting i) delete 10 ii) delete 14	10
5	CO5	Knowledge	Define directed graph. Illustrate two different methods to store a graph.	10

PART – B

Q. No	CO	Cognitive Domain	Question	Marks
6.a)	CO1	Analysis	Distinguish between static and dynamic allocation.	04
b)	CO1	Application	Write a 'C' function to delete an element in an array. After deleting an element array should not have any holes.	06
			OR	
7	CO1	Application	Develop a 'C' program to implement the structure named Book with members Book-id, Title, Author, Price. Write functions to i) Read ii) Display iii) Search for a book based on Book-id iv) List all books of a given author. Note: use static / dynamic allocation.	10

8	CO2	Analysis	List the drawbacks of simple queue. How these drawbacks are overcome in circular queue. Give the implementation of circular queues.	8+2=10
			OR	
9.a)	CO2	Application	Write the functions to implement the operations of input restricted Deques using arrays.	06
b)		Knowledge	Write a recursive function to solve Tower of Hanoi problem.	04

10	CO3	Application	Consider a singly linked list to represent a polynomial. Implement addition of two polynomials 'a' and 'b' and store the resultant polynomial in C, where a, b and c represent singly linked list.	10
			OR	
11	CO3	Application	Implement the operations using doubly linked list with header node. i) Insert-Begin ii) Insert-End iii) Delete-Begin iv) Delete-End v) Display	10

12	CO4	Knowledge Application	Define expression trees. Construct an expression tree for the expression i) $/*+AB+CD+*EFG$ ii) $AB-CD+*EF*GH/-$	10
			OR	
13.a)	CO4	Knowledge	Define i) Strictly Binary tree ii) Binary Search Tree	04
b)		Comprehension	Give the three traversals for the tree in fig. 1.	06

14.a)	CO5	Comprehension	Define priority queue. Explain different types of priority queues.	06
b)			Explain the graph traversal techniques.	04
			OR	
15	CO5	Application	Apply BFS and DFS on graph in fig. 2.	10

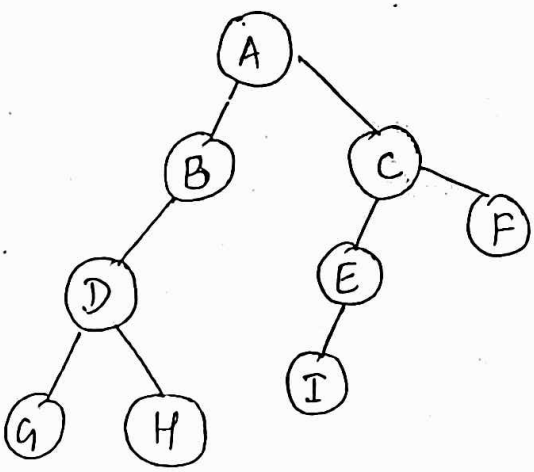


Fig. 1

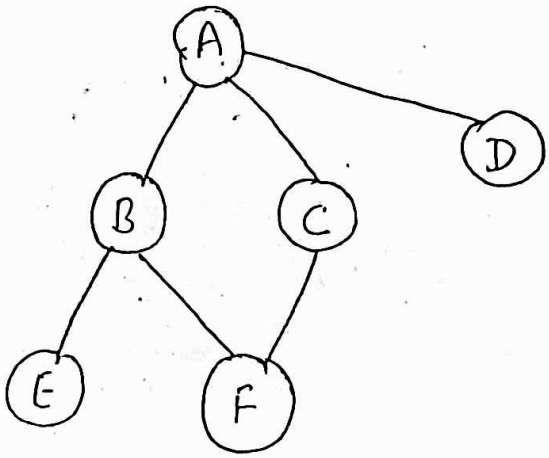


Fig. 2

CODE NO: CS340

JSS Mahavidyapeetha
JSS Science and Technology University
JSS Technical Institution's Campus,
Mysuru – 570 006

III Semester B.E. Degree
Semester End Examination, December 2018

Branch : Computer Science & Engineering Section: A, B & C

Name of the Paper Setter: Sheela N & Brunda S

DATA STRUCTURES

Duration: 3 Hours Time: 9.30 am to 12.30 pm Max. Marks: 100
Date: 12.12.2018 Day: Wednesday

Course Outcome:	
CO–1	Understand and analyze the fundamentals of data structures and their applications essential for problem solving.
CO–2	Implement various operations on linear lists.
CO–3	Analyze linear data structures stack, Queues and develop applications using the same.
CO–4	Comprehend the concept of non-linear data structure Tree and develop applications.
CO–5	Implement priority queues for various applications and use non linear data structure graph for different applications.

Instructions to the Candidate:
1. Part A is compulsory.
2. Part B has internal choices.

PART – A

Q. No	CO	Cognitive Domain	Question	Marks
1 (a)	CO 1	Knowledge	Define Data Structure. Describe how they are classified. Give example for each.	05
(b)	CO 1	Comprehension	Explain abstract data type with an example.	05

2	CO 2	Comprehension & Application	Discuss the advantages of a circular linked list over Non – circular linked list. Write a function to create an ordered singly linked list.	10
3	CO 3	Application	Write a routine to convert intix expression to its postfix form. Trace the algorithm on the following expression by showing the contents of the stack at each stage. (((A+B)$D) / (E – F) + G $ H)	10
4	CO 4	Application	Construct a Binary search tree (BST) for the given values 14, 5, 6, 2, 18, 20, 19, 16, -1, 21 and write a function to delete a node from a Binary search tree. Also draw the tree after deleting a node with values 18.	10
5	CO 5	Comprehension	Define Graph and Illustrate with example different storage representation of a graph for both directed and undirected graph.	10

PART – B

Q No	CO	Cognitive Domain	Question	Marks
6 (a)	CO 1	Analysis	Debate on the advantages and disadvantages of array representation over linked representation.	06
(b)	CO 1	Comprehension	List and explain with an example, the problems of not deallocating the dynamically allocated block of memory using free ( ).	04
			OR	
7	CO 1	Application	Develop a 'C' program to implement a structure called student with members like USN, name, marks 1,marks 2, marks 3, total. Write functions to i) Read details of 'n' students ii) To find the average of best two marks of each student iii) Search for the students details based on USN and display iv) To display the details of all the students use dynamic allocation for structure.	10
8 (a)	CO 2	Application	Design a function to delete all the nodes with information 'X' in a circular double linked list with header node, also print the number of nodes deleted else print suitable message.	05

(b)	CO 2	Application	Write a routine called ADD-POLY (P1, P2, P3) which adds two polynomials pointed by P1 and P2 and returns the result pointed by P3.	05
			OR	
9 (a)	CO2	Application	Design a function called search update (H, X, Y) that searches a node with information 'X' and updates it by 'Y' in a circular singly linked list with header node pointed by H, provided 'X' exists in the list.	05
(b)	CO2	Application	Design a function Reverse (F) to reverse a single linked list without creating new nodes, where 'F' points to the first node in the list.	05

10	CO 3	Application	Transform the following infix expressions into their equivalent prefix forms by tracing the stack contents. i) A * (B + D) / E – F * (G + H / K) ii) (A + B) * (C – D) $ E * F	10
			OR	
11 (a)	CO 3	Comprehension	Discuss the drawbacks of linear queue. How these drawbacks can be overcome in circular queue. Give implementation of circular queue.	06
(b)	CO 3	Application	Apply the required algorithm to find the value of the following prefix expression for A = 10, B = 2, C = 12 and D = 2, E = 2, F = 3 by tracing the contents of the stack. + / $ - A B E $ / - C D / A B F / + A C B	04

12 (a)	CO 4	Comprehension	What are the total number of nodes and total number of non – leaf nodes in a complete binary tree of depth 'd' prove the same.	05
(b)	CO 4	Application	Construct a Binary Tree for the given traversals showing the constructed tree at each step. Preorder : A B D G C E H I F Inorder : D G B A H E I C F	05
			OR	

13	CO 4	Application	Construct an expression tree for the following expressions and write the other two forms of the expression by traversing the constructed tree i) + A * - B C $ D * E F ii) A B C * + A B + C * $	10

14	CO 5	Application	For the graph in Fig 14.1 give the DFS and BFS traversals starting from vertex 'a' by tracing the contents of the required data structures. Use adjacency matrix representation.	10
			OR	
15	CO 5	Comprehension	What is a Heap? Illustrate with an example, implementation of priority queue using Binary Heap data structure.	10

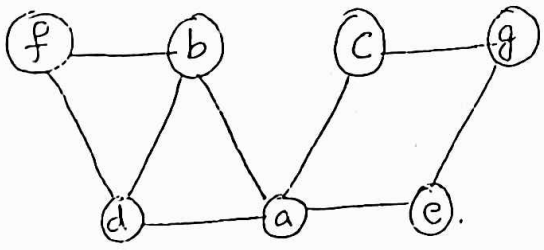


Fig - 14.1

JSS Mahavidyapeetha
JSS Science And Technology University
(Established Under JSS Science and Technology University Act No. 43 of 2013)
(Formerly Known as SJCE)

JSS SCIENCE AND TECHNOLOGY UNIVERSITY MYSURU

III Semester BE
Semester End Examination, DECEMBER 2019

Branch : Computer Science & Engineering **Section: A, B, C,**

Name of the Paper Setter: Dr. H. C. Vijayalakshmi. Sheela. N

CODE NO: CS340 **DATA STRUCTURES**

Duration: 3 Hours **Time: 9.30 AM to 12.30 PM** **Max. Marks: 100**
Date: 21.12.2019 **Day: Saturday**

Instructions to the Candidate:
Answer all Questions from PART – A.
PART – B has internal choice.

Course Outcome:	
CO–1	Understand the fundamentals of data structures and their applications for problem solving.
CO–2	Apply Linear and Non-Linear data structures to organize data for different types of problems.
CO–3	Analyze and implement various operations on Linear and Non – linear data structures.
CO–4	Evaluate the most appropriate data structures for efficient data storage for a given problem.

PART – A

Q. No	CO	Cognitive Domain	Question	Marks
1	CO-1	Knowledge Comprehension	a) What is data structures? Discuss the classification of data structures with an example for each. b) Discuss the two storage representation with their advantages and limitations.	05 05
2	CO-3	Analysis	Write routines to perform the following operations on circular single linked list with header node by defining the node structure. a) To insect a new node by position b) To delete immediate right node of Nth node.	 05 05
3	CO-2	Application	Convert the following expressions to other two forms by tracing the stack contents. a) (A – (B+C)) * D$(E + F/G$H)	10
4	CO-1	Knowledge Comprehension	a) Discuss with examples the various storage representations for tree data structure. b) What is a complete binary tree? What are the total number of nodes, leaf nodes and non-leaf nodes in a complete binary tree of depth 'd'.	05 05
5	CO-3	Analysis	Analyze the advantages and disadvantages of adjacency list and adjacency matrix for the representation of directed and undirected graphs with examples.	10

PART – B

Q. No	CO	Cognitive Domain	Question	Marks
6	CO-2	Application	a) Write a function called REVERSE that receives a pointer to an array of integers and number of elements exists in the array as arguments and reverses the contents of the array. b) Write a function called INSERT that receives a pointer to a sorted array of integers; no of elements and an integer value as arguments and inserts a value in its correct place.	05 05
			OR	
7	CO-2	Application	a) Write a function called COMPARE which receives pointers to two integer arrays and number of elements as arguments and compares the two arrays to check whether they are identical. The function returns 1 if they are identical, 0 otherwise. b) Write a recursive function to find the Nth Fibonacci number and trace the same by writing tree of recursive calls for N = 5	05 05
8	CO-4	Evaluation	Choose and explain with an example, an appropriate data structure to represent polynomials efficiently and implement the operation ADD which adds two polynomials.	10
			OR	

Q. No	CO	Cognitive Domain	Question	Marks
6	CO-2	Application	a) Write a function called REVERSE that receives a pointer to an array of integers and number of elements exists in the array as arguments and reverses the contents of the array. b) Write a function called INSERT that receives a pointer to a sorted array of integers; no of elements and an integer value as arguments and inserts a value in its correct place.	05 05
			OR	
7	CO-2	Application	a) Write a function called COMPARE which receives pointers to two integer arrays and number of elements as arguments and compares the two arrays to check whether they are identical. The function returns 1 if they are identical, 0 otherwise. b) Write a recursive function to find the Nth Fibonacci number and trace the same by writing tree of recursive calls for N = 5	05 05
8	CO-4	Evaluation	Choose and explain with an example, an appropriate data structure to represent polynomials efficiently and implement the operation ADD which adds two polynomials.	10
			OR	
9	CO-3	Analysis	Write routines to perform the following operations on circular double linked list with header node by defining the node structure. a) Search for a node with the given key value, delete if exist and append to the end. b) Delete predecessor of the Nth node.	06 04

10	CO-3	Analysis	a) Discuss the draw backs of ordinary Queue with an example. Explain any two methods to overcome the same.	06
	CO-2	Application	b) Discuss how categorization of data can be done maintaining their original sequence efficiently using Queue data structure.	04
			OR	
11	CO-3	Analysis	Write the following routines by defining a Queue structure with array storage. a) To insert items into a Queue in ascending order. b) To delete highest prioritized item in an unordered Queue.	05 05
12	CO-2	Application	Construct the binary search tree for the sequence 70, 15, 80, 19, 90, 25, 85, 75, 35, 78, 10, 29, 86, 18. Write different traversals. Illustrate how the following sequence of deletion operations are performed writing the resultant tree after each operation. i) Delete 15 ii) Delete 90 iii) Delete 70	10
			OR	
13	CO-2	Application	Construct the expression trees for the following expressions and write the other two forms of the expressions by traversing in the required order. i) * - A + B C $ D + E F ii) ((A + B) * C) $ D / E $ F / (G – H)	10
14	CO-4	Evaluation	Choose an appropriate data strcuture and explain with an example to represent adjacency relationship between the cities and write an algorithm to find the list of cities adjacent to each city.	10
			OR	
15	CO-3	Analyses	Write the algorithms for different graph traversals methods and trace the same for the graph shown in fig 15.1	10

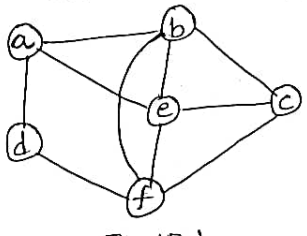


Fig 15.1